

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\lim_{r \rightarrow \infty} \int_{\mathcal{I}^-} dt d\phi dt' d\phi' \left[\frac{(rr')^2}{l^2} \left(\Phi(r, t, \phi) \overset{\leftrightarrow}{\partial}_{r_*} G(r, t, \phi; r', t', \phi') \overset{\leftrightarrow}{\partial}_{r'_*} \Phi(r', t', \phi') \right) \right]_{r=r'},$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\partial_a^m \Gamma_i = \frac{\Gamma^n}{\lambda_i} \{ \delta_{nm} \psi_a^i - \phi_b^n \phi_c^m \psi_b^i \psi_c^i \frac{\psi_a^i}{\lambda_i^2} + \phi_b^n \phi_c^m \sum_{j \neq i} \psi_b^j \frac{(\psi_c^i \psi_a^j + \psi_a^i \psi_c^j)}{(\lambda_i^2 - \lambda_j^2)} \}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\langle E_{eff}(x_0, x_1) \rangle = \frac{1}{6} \left[\frac{\partial^2}{\partial x_0^2} \frac{\partial^2}{\partial y_1^2} - \frac{\partial^2}{\partial x_0 \partial x_1} \frac{\partial^2}{\partial y_0 \partial y_1} \right] \Delta_\varphi(x_0, x_1; y_0, y_1)|_{y_\mu \rightarrow x_\mu},$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\frac{\delta}{\delta \phi_A^*} + \frac{i}{\hbar} \frac{\delta \Gamma}{\delta \phi_A^*} - (-1)^{\varepsilon_i(\varepsilon_A+1)} (\Gamma''^{-1})^{ij} \left(\frac{\delta l}{\delta \varphi^j} \frac{\delta \Gamma}{\delta \phi_A^*} \right) \frac{\delta l}{\delta \varphi^i},$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\hat{h}_{\text{red}} = \sum_{\vec{b}} \left\{ -\frac{g^2 LM}{4\sqrt{2}} \frac{1}{J} \frac{\delta}{\delta \varphi_{\vec{b}}} J \frac{\delta}{\delta \varphi_{\vec{b}}} + \left(\frac{4\sqrt{2}}{g^2 LM} \right) \sum_{\vec{\varepsilon}} (\varphi_{\vec{b}} - \varphi_{\vec{b}+\vec{\varepsilon}})^2 \right\}.$$